

DSAA3073: Theories in Data Science

Week 10 Open Investigation

What Should Online Gradient Descent Compete With?

A data-centre reserve policy under changing comparators and curvature

Basic analysis. Scheduling emergency cooling reserve

Use the OCO protocol, Euclidean projection, and OGD theorem from the Week 10 Learning Sheet

Every ten minutes a data-centre controller chooses a reserve fraction $x_t \in \mathcal{K} = [0, 1]$ before the next cooling demand $d_t \in [0, 1]$ is known. After the interval, the convex operating loss is

$$f_t(x) = \frac{1}{2}(x - d_t)^2.$$

The loss records both under-reserving and unnecessary reserve. For an illustrative three-interval audit, take $x_1 = 0.50$, step size $\eta = 0.50$, and realized demands $(d_1, d_2, d_3) = (0.90, 0.10, 0.80)$.

Basic Task 1. Execute the projected OGD policy

Derive the gradient and compute x_2, x_3, x_4 , explicitly showing the unprojected point and its Euclidean projection on every round. Calculate the controller's three losses. Explain why the update uses f_t only after x_t has already been committed.

Basic analysis. Certify performance against one fixed reserve

The theorem is sequence-wise and does not assume stationary demand

Over one day there are $T = 144$ intervals. On $[0, 1]$, the diameter is $D = 1$ and $|\nabla f_t(x)| \leq 1$, so take $G = 1$.

Basic Task 2. Tune and interpret the OGD regret bound

State the constant-step OGD theorem, choose the balanced legal step, and compute the resulting numerical bound. Identify the comparator precisely. Explain why the result can hold for an arbitrary realized demand sequence without implying a population-risk or forecasting guarantee.

Thinking investigation. Formalize the comparator contract

A small static-regret theorem need not track a changing operating optimum

Write the analysis as

$$(\mathcal{K}, (f_t)_{t=1}^T, \mathcal{C}, \mathcal{G}),$$

where $\mathcal{C}$ is the comparator class. The basic theorem uses $\mathcal{C} = \{(u, \dots, u) : u \in \mathcal{K}\}$, the set of constant comparator paths. Suppose operations instead care about a feasible path $u_1, \dots, u_T \in \mathcal{K}$ whose total movement is

$$P_T = \sum_{t=1}^{T-1} \|u_{t+1} - u_t\|_2.$$

Investigation 1. Replace the fixed comparator by a moving one

Does the Week 10 static-regret theorem directly control $\sum_{t=1}^T (f_t(x_t) - f_t(u_t))$? Starting from the one-step OGD inequality, derive a dynamic-regret upper bound in terms of D , G , T , η , and P_T . Optimize the bound over η and interpret the role of comparator variation.

Thinking investigation. Strengthen convexity of the losses

Additional curvature changes the analytic rate

Return to a fixed comparator u . Suppose now that every f_t is α -strongly convex and has gradient norm at most G . Projected OGD uses the decreasing step $\eta_t = \frac{1}{\alpha t}$.

Investigation 2. Determine whether the square-root rate is still the endpoint

Insert the strong-convexity inequality into the OGD proof and derive a logarithmic regret bound. Identify the cancellation that is unavailable for merely convex losses. Does a fixed strongly convex regularizer, with otherwise arbitrary convex losses, license the same conclusion?

Synthesis. State the assumption-guarantee map

Compare general convex losses against one fixed comparator, general convex losses against a comparator path of variation P_T , and strongly convex losses against one fixed comparator. State the corresponding analytic rate and the assumption responsible for each change.